Saurabh Kumar

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Summary

Full-Stack Developer and Computer Science undergraduate with hands-on experience building production web applications using React.js, Node.js, Express.js, FastAPI, and MongoDB/SQL. Skilled in designing scalable RESTful APIs, implementing robust authentication (OAuth 2.0, JWT), real-time communication via WebSockets, containerizing microservices with Docker on AWS EC2, and integrating AI models into user experiences.

Education

Dr. APJ Abdul Kalam Technical University — Lucknow, India
Bachelor of Technology in Computer Science and Engineering; CGPA: 7.26/10.0 — *Aug. 2023 – May 2027*

Uttar Pradesh Board (UPMSP) — Uttar Pradesh, India
Secondary School Certificate (Class X); Percentage: 87% — *May 2020*

Technical Skills

Programming Languages: JavaScript (ES6+), Python, C++, HTML5, CSS3, SQL
Frontend Development: React.js, Vite, Tailwind CSS, Redux Toolkit, Context API, Responsive Design, SPA
Backend & APIs: Node.js, Express.js, FastAPI, RESTful APIs, WebSockets (Socket.io), JWT, OAuth 2.0
Databases & ORM: MongoDB, MongoDB Atlas, MySQL, Mongoose
DevOps & Cloud: Docker, AWS EC2, Nginx, GitHub Actions (CI/CD), Vercel, Render, Git, Linux/Bash
AI Integration: LangChain, LangGraph, Groq LLM Function Calling, Google Gemini API, RAG Pipelines, Vector DBs
AI Tools: Codex, Antigravity IDE, Cursor
Core Competencies: Data Structures, Algorithms, System Design, Microservices, API Optimization

Projects

HavenTo – Accommodation Booking Platform | *React, Node.js, FastAPI, MongoDB, Docker, AWS EC2* — Mar. 2026

- Architected full-stack property booking platform supporting property discovery, reservation lifecycle, and multi-tier host management.
- Engineered an autonomous AI travel concierge using Groq LLM with function/tool calling (property lookup, booking queries) and RAG conversation memory.
- Built a dynamic pricing microservice with REST endpoints, exposing sub-150ms real-time rate recommendations to host dashboard with one-click adjustments.
- Containerized services with Docker and deployed to AWS EC2 behind an Nginx reverse proxy with SSL; established automated GitHub Actions CI/CD for testing and deployment.
- Implemented secure JWT authentication, OTP email verification, Socket.io real-time chat, and Cloudinary media pipelines with client-side image compression.
- Links: GitHub | Live Demo

StudyMate – AI-Powered Collaborative Workspace | *React, Node.js, Express, MongoDB, Gemini AI* — Oct. 2025

- Engineered full-stack educational platform with AI-assisted learning, automated document summarization (PDF/DOCX), and collaborative study rooms.
- Implemented an interactive Obsidian-style Knowledge Graph using 2D force-directed physics simulation with bidirectional markdown backlinks and automatic tag clustering.
- Developed student retention and study analytics dashboard tracking weekly hours, streak consistency, and inactivity signals with predictive retention scoring.
- Integrated Google OAuth 2.0 Single Sign-On, JWT session guards, and rate-limiting middleware for enterprise-grade security.
- Deployed distributed architecture on Vercel (frontend) and Render (backend) with MongoDB Atlas for high-availability data persistence.
- Links: GitHub | Live Demo

Brain Tumor Detection Portal | *Python, TensorFlow, Keras, Express.js, React, OpenCV* — May 2025

- Engineered full-stack medical diagnosis portal achieving 95.59% accuracy in detecting brain tumors from MRI scans using custom CNN with Transfer Learning (VGG16).
- Integrated an Express.js REST API with Python deep learning models via asynchronous child processes to stream image uploads and deliver instant inference results.
- Designed responsive hospital dashboard in React featuring secure license-based registration, Multer file uploads, and interactive confidence score visualization.
- Preprocessed and normalized 3,000+ brain MRI scan images using OpenCV and NumPy for model training, validation, and testing.
- Links: GitHub